NISR
NATIONAL INSTITUTE OF STATISTICS OF RWANDA

NISR 2026 Big Data Hackathon Preparatory Webinar

NISR Data Science Team

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Introduction to NISR

- NISR is responsible for the collection and publication of statistics related to the economy, population and society of Rwanda.
- NISRs mission is to assume the leading role in improving capacity to use information for evidence-based decision-making by coordinating national effort to collect and archive reliable data, to analyze, document and disseminate data within an integrated and sustainable framework.
- NISRs vision is to develop and sustain a culture of excellence in statistical production and management of national development.

Introduction to NISR Data Science

- The data science team at NISR was started in 2021
- Our mission is to scope and deliver data science and Big Data projects that support NISR to achieve its mandate of supporting evidence-based decision making across government

NISR Big Data Hackathon

- Aim of the hackathon is to raise awareness of data science for official statistics amongst students
- To foster evidence-based policy-making through innovative tools.
- Enhancing the use of national data in public and private decision-making and Identify emerging talents in Rwanda for data science
- Build data science capacity across academic institutions.

Requirement for Participation

- Competition is open to all undergraduate or master's students, regardless of their field of study.
- Students must have a **student card or other official document** confirming he/her eligibility.
- A team must be composed of 2 students with at least one Rwandan citizen. Team members can be from the same university or different universities or Colleges.
- Participants are allowed to use any programming language of their interest.

Requirement for Participation

- By entering the competition, participants agree to assign and transfer all intellectual property rights in their submitted work, including but not limited to copyright and patent, to the National Institute of Statistics of Rwanda (NISR). NISR reserves the right to use, reproduce, modify, publish, and distribute the submitted work in any form and for any purpose.

Submissions

- Submissions for the hackathon will be expected as a public GitHub link and a deployed app link and documentation
- Plagiarism is prohibited. You must declare that the submission is your own work and that all sources of reference are acknowledged in full. Failure to do so will result in disqualification.

Judging Criteria

1. Problem Understanding & Relevance

   Clarity in identifying and understanding the challenge and its alignment to national priorities (NST2 or vision 2050 alignment).

2. Data Use & Methodology

   Appropriate and innovative use of datasets (NISR + external), sound analytical or AI/ML methods, and data preprocessing quality.

3. Tangible Impact

   Potential for real-world use by government, Rwandan citizen, civil society, or development partners.

Judging Criteria

4. Tech Innovation

   Quality and relevance tools and technology used to develop the solution. Creativity in approach (e.g., predictive models, satellite analysis, smart alert systems).

5. Usability & Design of Tool/Prototype

   Intuitive interface, clear data visualization, or user-friendliness of the solution

How to Submit

- Submissions are expected by the **30 October 2026** in a public GitHub link and a deployed app link and documentation
- **deployed app link should:**

1. Be publicly accessible.
2. If login is required, provide sample credentials so users can log in easily.

How to Submit

- **This repository should have at least 4 files**:

1. A ReadMe file explaining what data your dashboard/application uses, what information it is trying to communicate, and how users can interact with what you're displaying.
2. A script(s) which contains the code for your solution/project which should work on any machine.
3. A video file (likely to be .webm or .gif) which shows you demonstrating the user interface and functionality of your solution/project, please use screen recording software to do this, Do not use your mobile phone to record the screen unless you are recording a mobile app or mobile-based solution that you developed. (screen record must not exceed 3 minutes)
4. A file containing the data you are using and a link to the data. If the data file(s) are too large, provide only the link and note this in your README file.

Screen Recording

- There are many options for screen recording using browser extensions
- Would recommend Screencastify in google chrome. You can watch how to use it here
- It let's you export video as a .gif or .webm file

Dashboard Tools - Shiny

- Main dashboarding tool used in R, already integrated with Rstudio
- More information at their website https://shiny.posit.co/
- Example applications found https://shiny.posit.co/r/gallery/
- An introduction to Rshiny is available on https://shiny.posit.co/r/getstarted/shiny-basics/lesson1/index.html, and the following textbooks are also useful:
- Interactive web-based data visualization with R, Plotly, and Shiny
- The Epidemiologist R Handbook – Chapter 43 Dashboards with Shiny

Dashboard Tools - Dash

- Dash is a dashboarding tool that works in R and Python. It is also linked to the Plotly visualisation library which can also be used in both languages.
- The dash website provides detailed documentation about how to use both Plotly and Dash. The app gallery has many example applications, this one has a simple demonstration of the functionality of Dash in Python.
- In addition to the resources on the Dash website, this YouTube series is very good at explaining how to get started.

Dashboard Tools - StreamLit

- StreamLit is a python based dashboarding tool
- As with Dash, StreamLit provide their own documentation on their website

GitHub

- GitHub is a web-based platform for hosting and sharing code.
- It helps teams collaborate on projects using version control (powered by Git).

Why GitHub

- Ensures your work is transparent, reproducible, and easy to share.
- Others can review both your code and documentation in one place.
- Supports teamwork: multiple people can contribute without overwriting each other's work.

Basic GitHub Workflow

- **Create a Repository:** On GitHub → “New Repository” → give it a name → add a short description.
- **Clone the Repository:** Copy the repo link → run `git clone <repo-link>` on your computer.
- **Add Your Work:** Place your project files (code, data, README, video, etc.) into the repo folder.

Basic GitHub Workflow

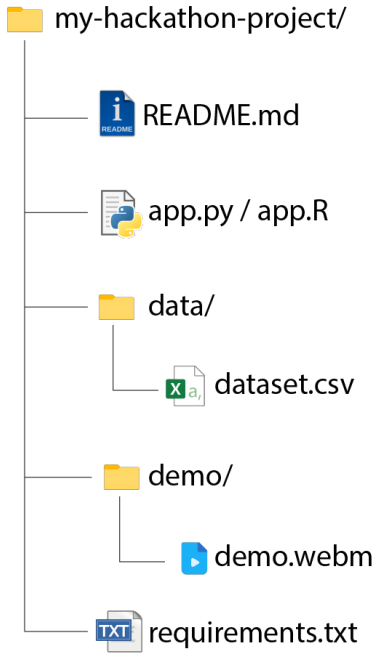
- **Commit Changes:**

```
git add .
git commit -m “Added initial project files”
```

- **Push to GitHub:**

```
git push origin main
```

Sample Repository Structure



GitHub Demonstration

- ✓ Create a GitHub account
- ✓ Create a repository for your project
- ✓ Create a ReadMe file
- ✓ Upload files to GitHub

GitHub Advanced

- What I've shown is a very basic way of using GitHub to store files
- Normally you would use GitHub continuously through your project, linking it to your personal computer and uploading directly (We will explore this in our next webinar)
- More information about how to do this can be found here.

Hints and Tips

- Think about using other datasets that could be relevant, for example shapefiles to make map based visualisations.
- Focus on functionality over fancy aesthetics, it's more important that we understand the story you want to tell with the data than how good it looks.
- Use your time wisely. There are only 9 days left until the submission deadline, so do not wait until the last minute to finish your work.
- Have fun! We want every participant not only to compete but also to learn new skills that they can apply in their future university studies and work life.

How to Access NISR Datasets

1. Visit the NISR Microdata Portal

https://microdata.statistics.gov.rw/index.php/catalog

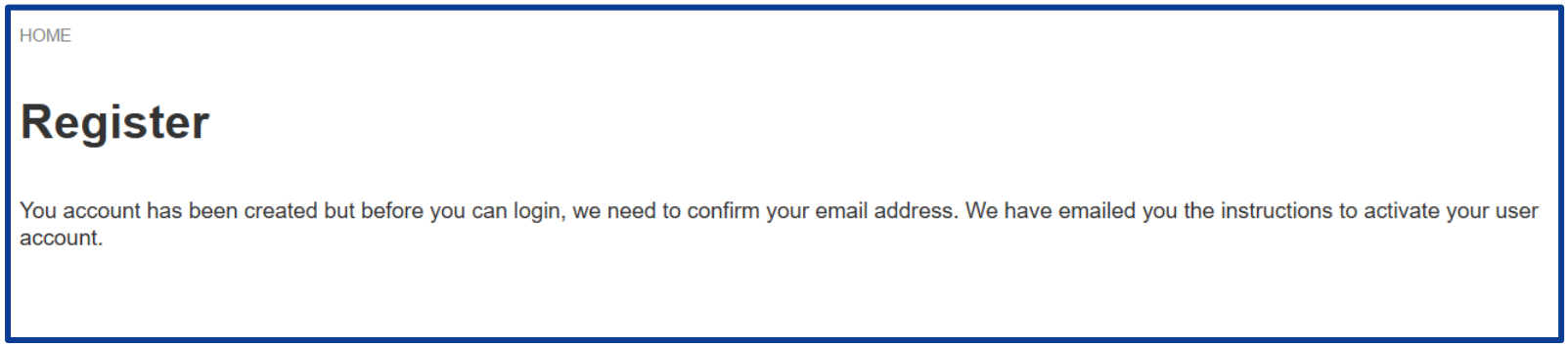
2. Create an Account

Click “**Register”** and fill in your details



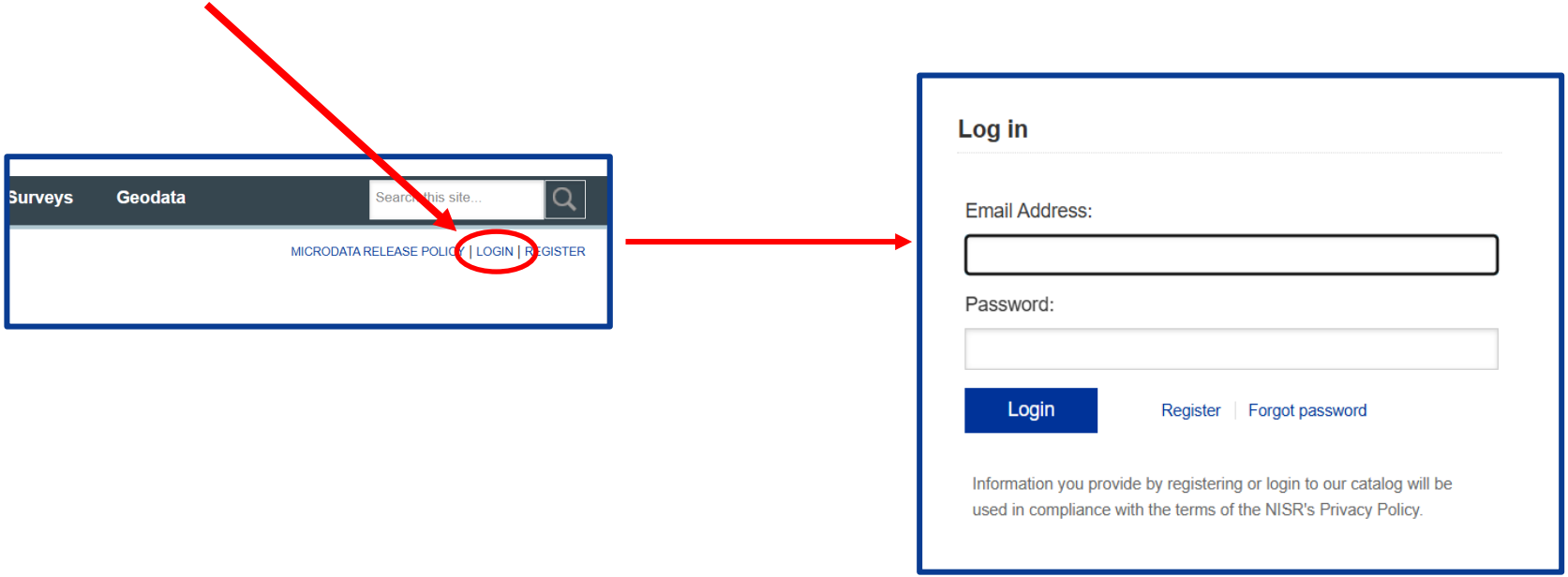
How to Access NISR Datasets

After registering, the portal will tell you a confirmation email has been sent. Ignore this



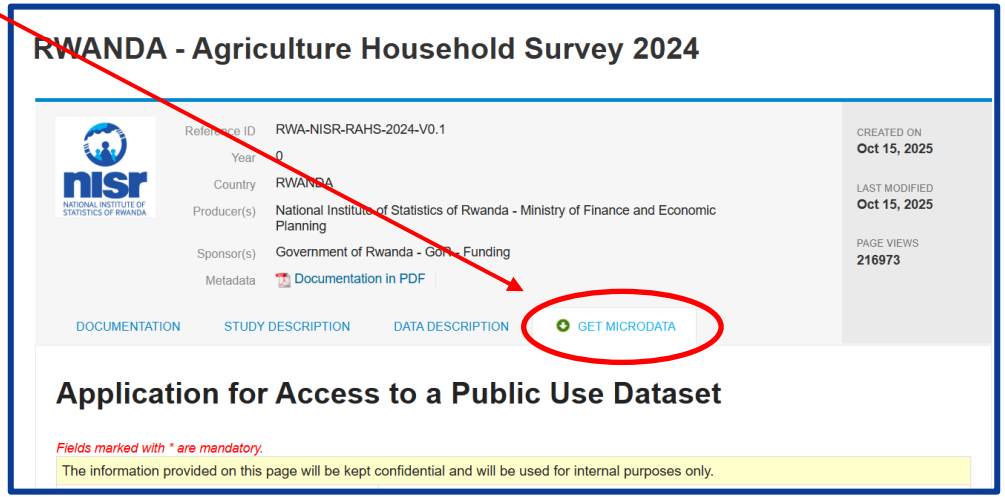
How to Access NISR Datasets

Simply log in directly using the credentials you set during registration



How to Access NISR Datasets

3. Browse & Search for Datasets
4. Click on your desired dataset
5. Click **“Get Microdata”** and submit the access request form



How to Access NISR Datasets

Access is usually granted immediately for public use datasets



Q & A

To keep things organized,
please write your questions in the chat.